

A MAJOR PROJECT FINAL REPORT ON

MEDICINAL PLANTS DETECTION STSTEM



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Project Work for the Degree of Bachelor's in Computer Engineering

MEDICINAL PLANT DETECTION SYSTEM

Supervised by Er. Sudeep Thapaliya, Er. Rajeev Poudel

A project work submitted in partial fulfillment of the requirements for the
degree of Bachelor's in Computer Engineering.

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DECLARATION

We hereby declare that this study entitled “**MEDICINAL PLANT DETECTION SYSTEM**” is based on our original work. Related works on the topic by other researchers have been duly acknowledged. We owe all the liabilities relating to the accuracy and authenticity of the data and any other information included hereunder.

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RECOMMENDATION

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ABSTRACT

Medicinal plants are foundational to global healthcare, particularly in regions like Nepal where traditional herbal medicine remains integral to daily life; however, manual identification is often complex and prone to human error, leading to the potentially unsafe application of herbal remedies. To address this, this project develops an automated Medicinal Plant Detection System leveraging the state-of-the-art YOLOv8 deep learning architecture. The system processes leaf images by utilizing a CSP Darknet backbone and C2f modules to automatically extract high-dimensional visual features, offering superior real-time performance and accuracy compared to traditional manual feature engineering or older CNN models. By providing an end-to-end solution that identifies species and details their medicinal properties, this tool serves as an accessible resource for students, researchers, and local communities using mobile or desktop interfaces. Ultimately, this research provides a scalable, high-speed deep learning solution that enhances the safety of herbal resource utilization and contributes significantly to the preservation of traditional botanical knowledge.

Keywords: *Medicinal Plants, Deep Learning, ResNet-50, Convolutional Neural Network, Plant Identification, Computer Vision, YOLOv8, Leaf Image Classification*

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ABBREVIATION AND ACRONYMS

CNN	Convolutional Neural Network
CPU	Central Processing Unit
GPU	Graphics Processing Unit
HSV	Hue, Saturation, Value (color space)
IoU	Intersection over Union
ResNet-50	Residual Network with 50 layers
ReLU	Rectified Linear Unit
RGB	Red Green Blue
ROC	Receiver Operating Characteristic
YOLO	You Only Look Once

CHAPTER 1: INTRODUCTION

1.1 Background

For generations, medicinal plants have been the backbone of healthcare in Nepal, bridging the gap between ancient local remedies and modern medicine [1]. In many of our communities, this herbal wisdom is a living heritage passed down from elders. However, there is a significant catch: identifying these plants correctly requires an expert eye. For an everyday user, a simple mistake in identification isn't just a technical error, it can lead to ineffective treatments or, worse, serious health risks from toxic look-alikes [2].

Fortunately, the rise of computer vision has changed the game. Instead of relying solely on human memory, we can now use deep learning to identify plants with high precision [3]. Modern frameworks like YOLOv8 (You Only Look Once) allow computers to analyze a leaf's shape, texture, and unique patterns in real time, [4]. Unlike older methods that required manual feature extraction, YOLOv8 learns these features directly from raw images, making the process faster and more reliable.

In this project, we have adopted the YOLOv8 architecture for our detection system. By utilizing its advanced backbone and C2f modules, the model can extract complex features from leaf images even in real-time scenarios [5]. This system is designed to be a practical and user-friendly tool for students, researchers, and local villagers, helping to preserve traditional medicinal knowledge in a digital and accessible format [6].

The proposed system is suitable for real-world applications and can assist students, researchers, and local communities in identifying medicinal plants using computers or mobile devices, while also supporting the preservation and digital sharing of indigenous medicinal knowledge [7].

1.2 Problem Statement

- Traditional plant identification relies on experts, which can be inaccurate for non-experts.
- Manual identification is slow and inefficient which leads to time consuming
- Misidentifying plants can lead to health risks, as some plants are toxic

1.3 Objectives

- To collect and prepare a dataset of medicinal plant leaf images suitable for deep learning training.
- To develop and fine-tune a YOLOv8-based model that achieves high accuracy in identifying various medicinal species.
- To design a simple and user-friendly interface that allows users to identify medicinal plants from leaf images and view basic medicinal information.

1.4 Scope and Limitation

Scope:

- The project focuses on leaf-based recognition; it does not handle roots, flowers, or full-plant images.
- We utilize the YOLOv8 model to ensure the system is not only accurate but fast enough for real-time applications.
- The system is designed for educational, research, and community use rather than commercial deployment.
- The output will include both species name and basic medicinal use (if data is available).

Limitations

- Performance may degrade with poor-quality images (blurry, low light, or cluttered backgrounds).
- Seasonal variation in leaves (such as changes in color and shape) may reduce identification accuracy.

- The system's species coverage depends on the available training data; rare or region-specific medicinal plants may not be included.
- Since only leaves are used for identification, species that require flowers, fruits, or other plant parts for accurate classification may be misclassified.
- The project does not detect or classify flowers, as it is specifically designed to identify plants based only on leaf characteristics.

CHAPTER 2: LITERATURE REVIEW

Medicinal plant identification using leaf images has gained significant attention due to its importance in healthcare, biodiversity conservation, and the preservation of traditional medicinal knowledge. Early research in this field primarily relied on classical machine learning techniques combined with handcrafted feature extraction methods. However, recent advancements have shifted the research focus toward deep learning approaches to improve accuracy and robustness.

Initial studies concentrated on extracting morphological, color, and texture features from leaf images. Morpho-colorimetric image features combined with spectroscopy data were classified using artificial neural networks (ANNs), achieving classification accuracy of up to 98.3% using image features alone [8]. A Group Labelled Classification (GLC) model utilizing both front and back morphological features of leaves was later introduced, resulting in improved recognition performance for medicinal plant species [9].

Texture-based feature extraction methods were also widely explored. Morphological features combined with Gray Level Co-occurrence Matrix (GLCM) texture descriptors were classified using ANN models, achieving an accuracy of approximately 87% for medicinal plant leaf recognition [10]. GLCM texture features were further integrated with hybrid Manhattan distance and Support Vector Machine (SVM) classifiers to identify herbal plant leaves, demonstrating strong classification performance [11]. Comparative studies analyzing texture operators such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Speeded-Up Robust Features (SURF) showed that LBP combined with SVM achieved very high accuracy for plant species classification [12].

Traditional machine learning classifiers such as SVM, K-Nearest Neighbors (KNN), and Random Forest were commonly used in these approaches. GLCM and color features combined with SVM were applied for leaf disease detection, achieving an accuracy of 98.79% in cross-validation experiments [13]. Although these methods produced satisfactory results, they required extensive manual feature

engineering and were sensitive to variations in lighting conditions, scale, and background complexity.

With improvements in computational power and the availability of larger datasets, deep learning has emerged as a more effective solution for plant identification tasks. Convolutional Neural Networks (CNNs) automatically learn discriminative features from raw images, eliminating the need for manual feature extraction. A CNN-based medicinal plant classification system trained on a self-curated dataset achieved accuracy exceeding 99% [14]. In another study, the VGG16 deep learning architecture was applied for medicinal plant recognition and reported an accuracy of 95.48%. Recent survey studies further emphasize the superiority of deep learning approaches over traditional machine learning methods, particularly in handling complex variations in leaf shape, texture, and environmental conditions.

2.1 Case Study

2.1.1 Indonesian Medicinal Leaf Classification Using Transfer Learning

A study conducted using the Indonesian Herb Leaf Dataset 3500 employed 3,500 leaf images distributed across 10 medicinal plant classes for classification tasks. Multiple transfer learning models were evaluated, with MobileNet demonstrating strong performance by achieving an accuracy of approximately 98.86%. The results highlighted the effectiveness of lightweight deep CNN architectures for medicinal plant identification, even when trained on relatively small datasets [15].

2.1.2 Medicinal Leaf Dataset Construction and Validation

Another study introduced a comprehensive medicinal leaf image dataset containing 11,040 samples collected from six medicinal plant species. The dataset consisted of 1,380 original images and 9,660 augmented samples to improve data diversity. The primary focus of the study was dataset construction, preprocessing, and validation rather than classifier evaluation; therefore, no specific classification accuracy was reported. Nonetheless, the dataset has been shown to support high-performance deep learning models in subsequent medicinal plant identification research. [16]

2.1.3 Efficient Net-Based Medicinal Plant Recognition

In a separate investigation, deep learning was applied to the Indonesian Herb Leaf Dataset 3500 using the EfficientNetV2B0 architecture. The proposed model achieved a classification accuracy of 99.14%, further reinforcing the effectiveness of modern deep learning architectures and transfer learning techniques for high-accuracy medicinal plant leaf recognition [17].

2.2 Summary of Related Works

Salsabila et al. utilized the Indonesian Herb Leaf Dataset 3500, consisting of 3,500 images from 10 medicinal plant classes, and achieved approximately 98.86% classification accuracy using the MobileNet model, demonstrating strong performance even with a relatively small dataset. Matin et al. introduced a medicinal leaf image dataset containing 11,040 samples from six plant species, focusing primarily on dataset construction and validation rather than classifier evaluation; therefore, no specific accuracy was reported, though the dataset serves as a valuable benchmark for training high-performance deep learning models. Similarly, Putra et al. applied the EfficientNetV2B0 architecture on the Indonesian Herb Leaf Dataset 3500 and reported a high classification accuracy of 99.14%, reinforcing the effectiveness of modern convolutional neural networks and transfer learning for medicinal plant leaf recognition.

2.3 Status of Medicinal Plant Detection System in Nepal

Nepal hosts a remarkably rich ethnomedicinal heritage, with over 1,700 documented medicinal plant species used across its diverse communities [1]. Despite this wealth of traditional knowledge, systematic efforts to detect and identify these plants using modern technology are still emerging. Recent ethnobotanical studies map out geographic hotspots of plant use particularly in the mid-hills and mountainous regions highlighting both conservation needs and gaps in documentation [18]. Concurrently, machine-learning approaches are beginning to be applied to Nepal's flora: a very recent study demonstrated a CNN-based mobile application capable of recognizing 60 different Nepalese herb species, showing promising accuracy and scalability [19]. However, much of the current

research is still focused on cataloging and traditional use rather than large-scale automated detection, indicating that Nepal is at the cusp of integrating AI-driven identification tools into its traditional medicine and conservation practices.

CHAPTER 3: METHODOLOGY

Methodologies of project management include a set of logically related practices, methods, and processes that determine how to plan, develop, control, and deliver projects throughout their continuous implementation process, until they are successfully completed and terminated. Design, execution, and completion of projects should be carried out according to this scientifically proven, systematic, and disciplined approach. Through effective decision making and problem solving, project methodology can ensure that processes, approaches, techniques, methods, and technologies are successful across the entire management process. To deliver and implement work according to the schedule, budget, and specifications of the client, a project management methodology provides a skeleton for describing each step-in depth. In the proposed methodology there are 4 different stages which are flow of project, system design, software and hardware requirement and testing and maintenance.

3.1 Medicinal Plant Detection System

This project is an automated medicinal plant identification system that utilizes the YOLOv8 classification model to recognize various plant species from leaf images. The system processes raw image data through specialized augmentation and training pipelines to ensure high prediction accuracy. The figure shows an overview of the medicinal plant identification and knowledge retrieval workflow.

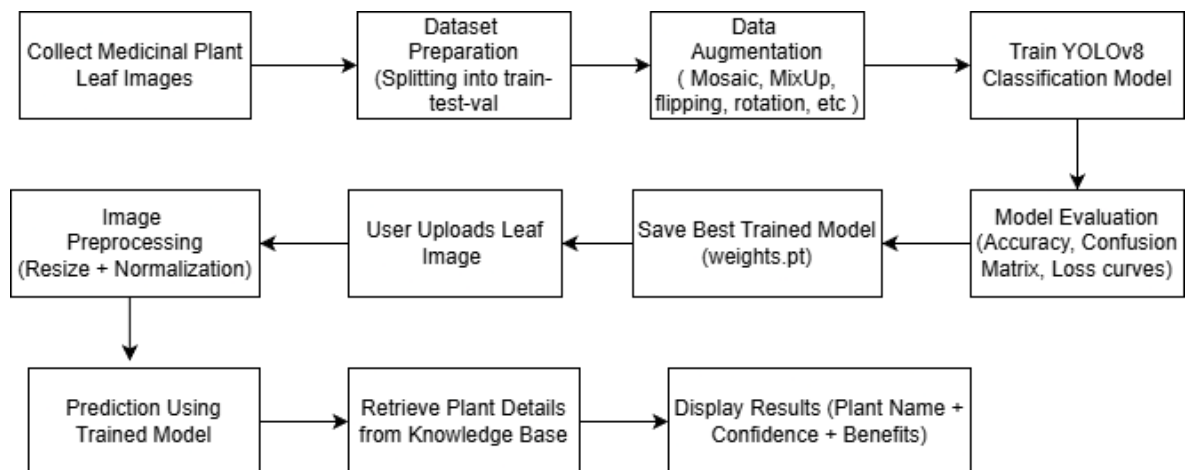


Figure 3. 1 : Generalize flowchart of System

3.1.1 Dataset Collection

- Organized the dataset into the standard YOLO classification format (separate folders for each class) to ensure compatibility with the Ultralytics training pipeline.
- Utilized a dataset containing at least 33 species of medicinal plants.
- Augmented data by collecting more images (if possible) from field photography or online sources (local medicinal plants).

3.1.2 Dataset Splitting

The dataset used in this project consists of 6,389 leaf images of 32 medicinal plant species, prepared for image classification using YOLO. Each image is labeled with its corresponding plant species, and no bounding boxes are required since this is a classification task. The images include variations in lighting, background, and leaf orientation, helping the model generalize well to real-world conditions. All images were preprocessed and resized for consistency during training. The dataset is split into training, validation, and testing sets (typically 70-15-15), with a distribution of images across classes, enabling the development of an accurate and robust model for medicinal plant identification.

3.1.3 Image Preprocessing

YOLOv8 handles much of the preprocessing internally, but the following pipeline was established:

- Resize all images to 224×224 pixels.
- We leverage YOLOv8's built-in mosaic and mix up augmentations, alongside standard rotations and flips, to increase dataset diversity.
- Images are normalized and converted to tensors automatically by the YOLOv8 data loader, eliminating the need for manual feature extraction.
- Data Augmentation: To improve the robustness and generalization of our model, we applied data augmentation techniques, specifically Mosaic and Mix-up augmentations. Mosaic augmentation combines four different images into one during training, allowing the model to learn from varied object scales and contexts within a single image. Mix-up augmentation blends two images and their corresponding labels, creating intermediate samples that encourage the model to learn smoother decision boundaries and reduce overfitting. By incorporating these techniques, the model is exposed to more diverse training scenarios, which enhances its ability to accurately detect and classify medicinal plant species under different condition

3.1.4 Training Setup and Configuration

For model training, high computational resources were required. Instead of setting up a local environment, Google Colab was used, which provides free access to NVIDIA GPUs with CUDA and cuDNN pre-installed. This eliminated the need for manual driver installation and environment configuration. The training was performed in a GPU runtime, ensuring faster processing and efficient model convergence [20].

3.1.5 Model Training

- Split the dataset into training, validation, and testing sets (70:15:15).
- We utilized YOLOv8-classification pre-trained weights. This allows the model to leverage features learned from the massive ImageNet dataset. Fine-tune the last few layers for medicinal plant classification.

- The model is trained using the yolo classify train command, optimizing hyperparameters such as learning rate, batch size, epochs and patience for early stopping.

YOLOV8 Architecture

YOLOv8 employs a CSP (Cross Stage Partial) backbone and a C2f module for more efficient feature extraction. It uses a "bottleneck" approach that combines high-level semantic information with low-level geometric features.

The classification head replaces the detection head found in standard YOLO models, producing a probability distribution across the 33 medicinal plant species. This architecture is specifically designed to be "anchor-free" and highly efficient, providing a significant boost in inference speed compared to ResNet-50 while maintaining high top-1 and top-5 accuracy.

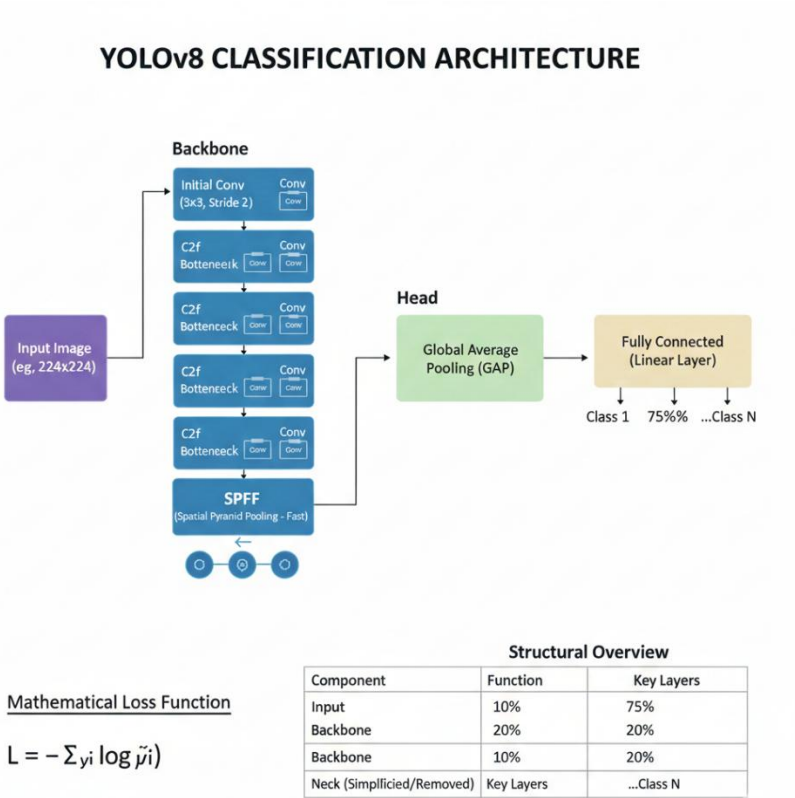


Figure 3.2: YOLOV8 Classification Model Architecture

Cross-Stage Partial Bottleneck with 2 Convolutions" that helps the model capture complex leaf textures more efficiently, which is why it outperformed ResNet-50 in your tests The C2f module is a Convolutional layer with Cross-Stage Partial (CSP) and "fuse" structure used in YOLOv8 to improve feature extraction while reducing computation. It's an evolution of the C3 module from earlier YOLO versions.

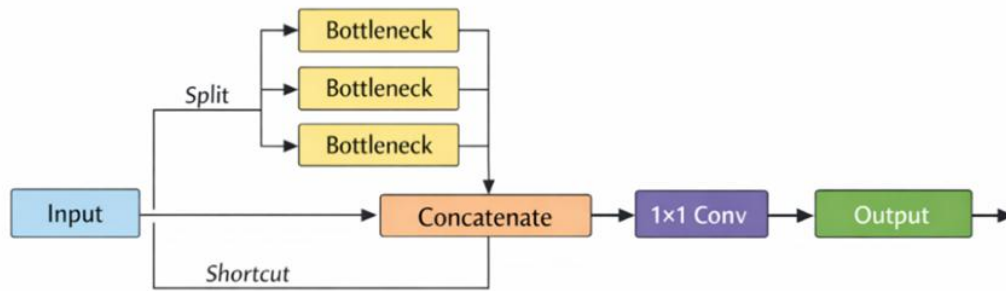


Figure 3.3: YOLOv8's C2f module

Comparative Model Analysis (YOLOv8 vs ResNet50)

In this project, we trained ResNet50 and YOLOv8 classification models on the same medicinal plant dataset containing 6389 images of 32 species. ResNet50 achieved stable performance after around 25 epochs, and training for 100 epochs did not significantly improve the results because the accuracy had already reached a plateau. However, when the same dataset was trained using YOLOv8 classification, the model achieved much stronger performance with very high Top-1 accuracy (~99.76%) and Top-5 accuracy (100%), along with very low validation loss. Therefore, YOLOv8 was selected over ResNet50 because it provided better accuracy, better generalization, and is also more suitable for future extension of the project (for example, upgrading from classification to object detection).

Table 3. 1 : Comparative Model Analysis (YOLOv8 vs ResNet50)

Feature	ResNet50 (CNN Model)	YOLOv8 (Classification Model)
Dataset Used	6,389 images, 32 species	6,389 images, 32 species
Epochs Trained	25 (also tested 100)	100
Validation Accuracy	~90% (from graph)	99.76% (Top-1)
Training Loss Trend	Decreases, then plateaus	Decreases strongly
Validation Loss Trend	Stable after ~25 epochs	Very low (~0.013)
Overfitting Risk	Moderate (accuracy saturates early)	Low (validation loss very low)
Overall Performance	Good	Excellent
Future Scalability	Mainly classification	Can be extended to detection easily
Final Choice	Not selected	Selected

3.1.6 Model Evaluation

The YOLOv8 classification model was evaluated using accuracy-based metrics, specifically Top-1 and Top-5 accuracy, as the framework reports these measures instead of precision, recall, and F1-score. Top-1 accuracy represents the percentage of samples where the model's highest-confidence prediction matches the true class, while Top-5 accuracy indicates how often the correct class appears within the model's five most confident predictions, making it suitable for multi-class problems with visually similar classes. In addition, a confusion matrix was used to examine class-wise prediction performance and identify common misclassifications. To assess learning behavior and detect overfitting, training and validation loss and accuracy curves were visualized, where closely aligned curves indicate good generalization and stable model performance.

3.1.7 Prediction & Deployment

- Save the trained YOLOv8's best.pt model for future inference.
- Develop a simple prediction interface (lightweight web GUI) for users to upload leaf images and receive predicted species.

- Test predictions on unseen images to verify real-world performance.

3.1.8 Documentation and Reporting

- Document code, dataset pipeline, and model experiments.
- Prepare a final report or mini research paper summarizing methodology, results, and findings.
- Create a presentation for project demonstration.

3.2 Development and Planning

3.2.1 Concept and Initiation

Before beginning the Medicinal Plant Identification System, it was important to study the problem and confirm that the project requirements were realistic and achievable. Since many medicinal plants have similar leaf structures, the system needed to correctly identify species using image-based classification. During this phase, the key requirements were gathered, such as building a model that works from leaf images, selecting a suitable deep learning approach, and ensuring the system provides meaningful output like plant name, confidence score, and medicinal benefits. This stage helped in understanding the scope of the project, choosing YOLOv8 classification as the core model, and preparing a clear plan for dataset collection, training, and system implementation.

3.2.2 Definition and Planning

In this Medicinal Plant Detection and Identification project, the main goal is to develop an AI-based system that can classify medicinal plants from leaf images accurately and efficiently. The system is designed to include important components such as dataset preparation (collection, cleaning, and splitting into training and testing sets), applying data augmentation techniques like Mosaic and MixUp to improve generalization, training the YOLOv8 classification model, and evaluating performance using accuracy, confusion matrix, and loss curves. After training, the best model is saved and integrated into a user interface where users can upload an image and receive the predicted plant name along with confidence score and

medicinal benefits retrieved from a knowledge base. This planning phase ensured that both the machine learning pipeline and the user-facing system were structured clearly for implementation.

3.2.3 Technology Used

- **Programming Language:** We used Python for model development, image preprocessing, and implementing the YOLOV8 architecture. Python provides flexibility, easy integration with deep learning libraries, and efficient handling of large image datasets.
- **Libraries / Frameworks:** Ultralytics (YOLOv8), OpenCV, PyTorch, NumPy, and Matplotlib. The Ultralytics framework simplifies the training process, while OpenCV handles real-time image processing.
- **Development Platform:** Google Colab was utilized for its GPU acceleration (T4/A100), which is essential for the rapid training and validation cycles inherent in YOLOv8's architecture.
- **Front-End / Interface:** HTML, CSS, and JavaScript provide a responsive interface for uploading leaf images. The backend serves the trained YOLOv8 .pt (PyTorch)

3.2.4 Launch and Execution

Once the development of the Medicinal Plant Detection System is complete, the system will be deployed for real-world usage. Users will be able to access the application to capture or upload leaf images and obtain accurate identification of medicinal plants along with relevant information. The system's compatibility across supported platforms will be ensured, and thorough testing will be conducted to validate its performance, accuracy, and reliability under different conditions. User interactions and feedback will be collected to identify potential issues and enhance system functionality, usability, and overall effectiveness through continuous improvements.

3.2.5 Performance and Control

The performance of the Medicinal Plant Detection System will be continuously monitored to evaluate its accuracy, response time, stability, and scalability. Regular performance evaluations and testing will be conducted to ensure that the system meets the defined project objectives and delivers reliable plant identification results. Proper control mechanisms, including secure data handling, model integrity checks, and regular backups, will be implemented to protect system data and maintain consistency. These measures will help ensure system reliability, security, and a smooth user experience during continuous operation.

3.3 System Design

The provided system flow diagram outlines a comprehensive, end-to-end pipeline for a medicinal plant identification system. The process initiates at the Start node and moves immediately into the User Interface, where a user uploads an image of a plant for analysis. Once the image is received, it undergoes a Preprocessing phase specifically resizing and normalization to ensure the data is in a consistent format for the neural network. The core of the system's computer vision capability lies in the YOLOv8 Feature Extraction layer, which utilizes the C2f module and the model's Backbone to identify intricate patterns and spatial characteristics within the image. Following feature extraction, the system enters the Classification stage, employing a SoftMax function to generate a Probability Distribution that determines the most likely species of the plant. Once the plant is identified, the system performs a Knowledge Base Lookup to retrieve detailed information regarding its Medicinal Benefits. Finally, the pipeline concludes at the Display Results stage, where the user is presented with a holistic output including the Plant Name, its therapeutic properties, and an interactive AI Chatbot Response for further inquiries, before looping back or terminating at the final Start/End node.

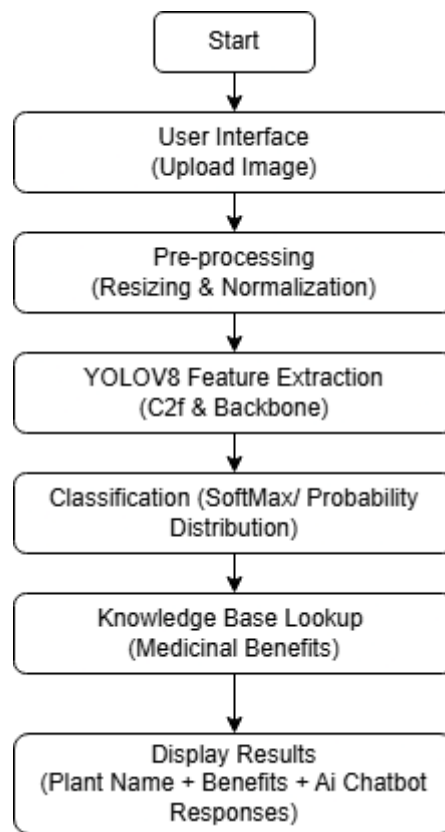


Figure 1.4: System Design Diagram

CHAPTER 4: RESULT AND DISCUSSION

4.1 Overview

The results show that the ResNet-50 model was able to identify medicinal plants from leaf images with good validation accuracy (~90%), demonstrating the usefulness of deep learning for plant recognition. Proper preprocessing and data augmentation helped the model learn meaningful features and make stable predictions, but the accuracy reached a plateau after around 25 epochs, with no major improvement even when trained up to 100 epochs. The model also faced difficulty with blurry images, complex backgrounds, and plant species having very similar leaf patterns. When the same dataset was trained using YOLOv8 classification, the performance improved significantly, achieving ~99.76% Top-1 accuracy with very low validation loss, showing better generalization and reliability. Overall, YOLOv8 was selected as the final model because it provided higher accuracy and stronger robustness for real-world use cases.

4.1.1 Training Performance and Metrics

The YOLOv8 classification model was trained on a dataset containing 6389 leaf images of 32 medicinal plant species. The model was trained for 100 epochs, and performance was evaluated using training loss, validation loss, Top-1 accuracy, and Top-5 accuracy. The results show that the model achieved strong learning convergence with high classification accuracy and low validation loss, indicating good generalization performance.

- **Metric Analysis:** The figure below illustrates the decrease in training and validation loss over time, alongside the increase in Top-1 and Top-5 accuracy.

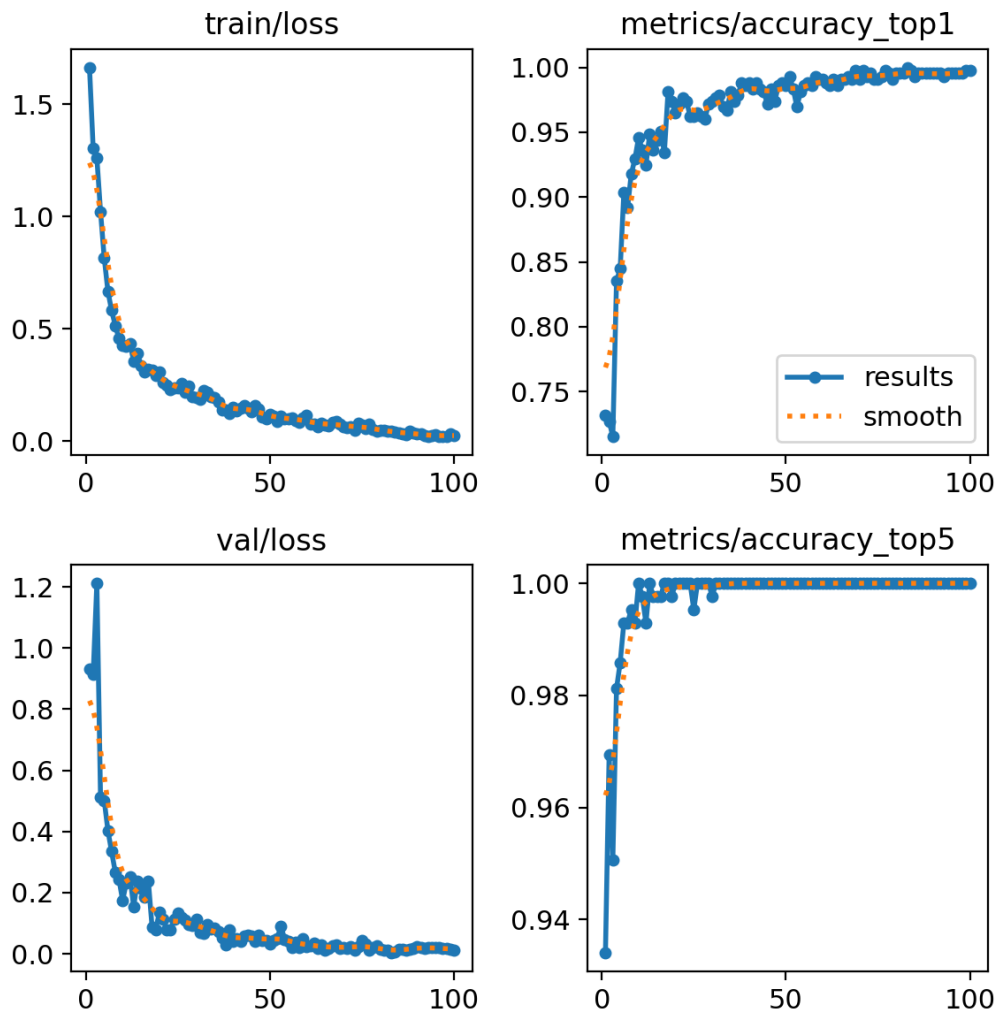


Figure 2.1: Training and Validation Metrics

4.1.2 Classification Accuracy and Confusion Matrix

To understand how well the model distinguishes between different species of medicinal leaves, we examine the confusion matrix.

- **Confusion Matrix:** The figure confusion matrix normalized provides a heatmap of predicted vs. actual classes. High values along the diagonal indicate strong predictive power. Any off-diagonal values represent "confusion" between leaf species with similar morphological features (e.g., similar vein patterns or shapes).

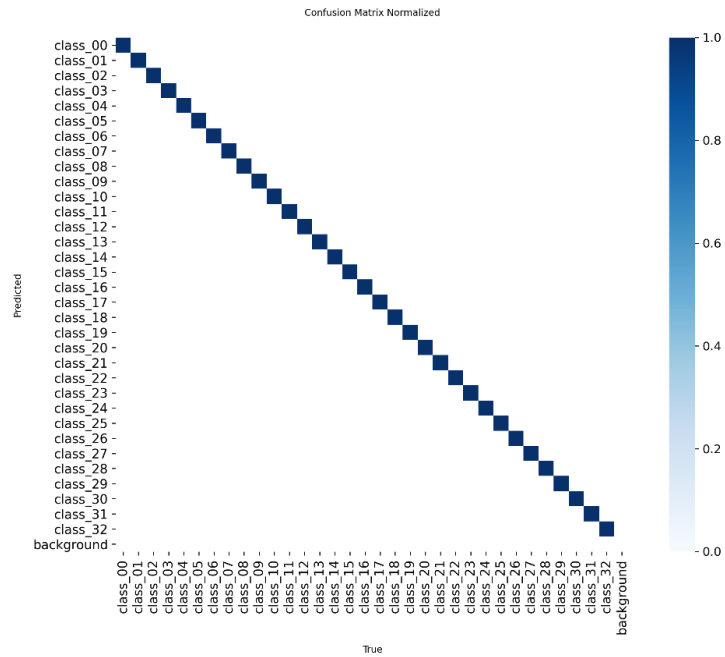


Figure 4.2: Normalized Confusion Matrix

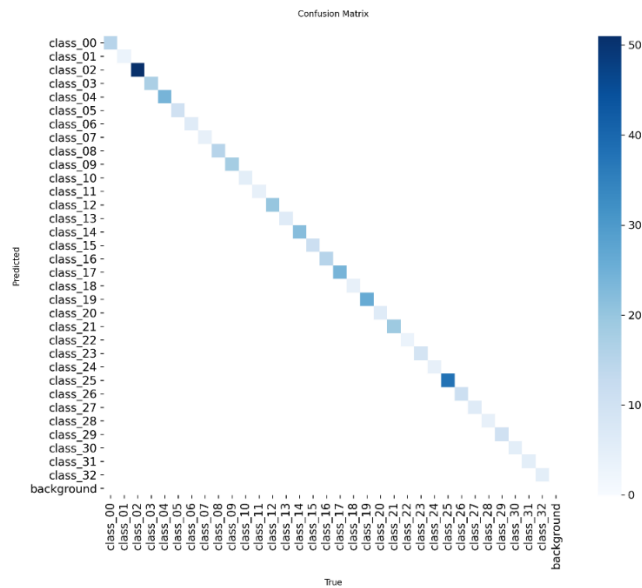


Figure 4.3: Confusion Matrix

The figure Confusion matrix (Standard Confusion Matrix) displays the raw count of predictions. The strong blue diagonal line indicates that the model is making correct predictions for most classes, as the "Predicted" labels match the "True" labels. The varying shades of blue on the diagonal show that some classes (like

class_02 or class_25) have more data samples than others, while the empty white space off the diagonal suggests there are very few misclassifications (False Positives or False Negatives).

The second image (Normalized Confusion Matrix): This scales the data so that each row or column represents a proportion (usually from 0 to 1). By normalizing the data, the impact of class imbalance is removed. In this view, the diagonal is a solid, dark blue across all classes, confirming that the model has a very high accuracy for every single category, regardless of how many samples that category originally contained.

4.1.3 Visualizing Data Augmentation and Model Predictions

A qualitative assessment was performed by inspecting the training batches and validation predictions.

- **Training Batches:** Figure like train_batch0 show the various augmentations (rotation, scaling, mosaic) applied to the medicinal leaf images, which help the model generalize better.
- **Validation Predictions:** The validation batch labels vs validation batch prediction figure allow for a side-by-side comparison of the ground truth labels against the model's actual predictions.



Figure 4.4: Train batch showcasing image augmentations



Figure 4.5: Ground Truth Labels and Model Predictions

4.1.4 Prediction

As seen in individual prediction results, the model assigns a confidence score to each leaf classification. High confidence scores (e.g 90% or above) for correct classes demonstrate the model's robustness in identifying medicinal species.

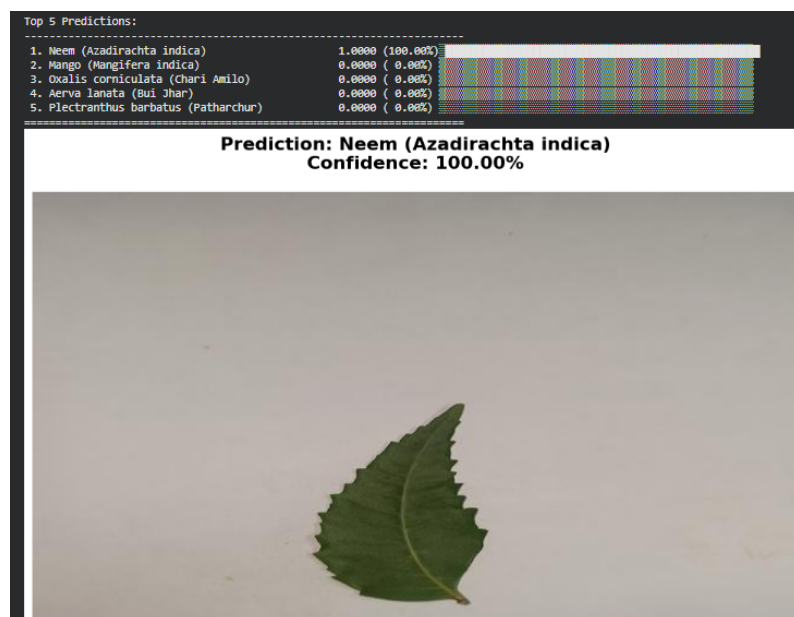


Figure 4.6: Classification Probabilities for the Input Plant Image

The YOLOv8-cls model proved effective for this medicinal leaf dataset, likely due to its advanced backbone architecture and effective data augmentation strategies like Mosaic augmentation seen in the train batch images. These results suggest that the model is suitable for real-time identification tasks where high speed and accuracy are required for herbal recognition.

Table 4. 1 : Table Showing 5 prediction results

S.N.	Predicted Plant	Scientific Name	Confidence
1	Gurjo	Tinospora cordifolia	100%
2	Guava	Psidium guajava	100%
3	Mango	Mangifera indica	100%
4	Aloe vera	Ghiu Kumari	100%
5	Mango	Mangifera indica	99.98%

When we evaluated the model on the images used for training, it predicted the top plants with 100% confidence. However, when we tested it on raw, unseen images, the confidence values were slightly lower (around 99%), which still indicates very reliable predictions. This shows that the model generalizes well to new data.

4.1.2 Application Interface and User Experience

The figure illustrates the user interface of the Medicinal Plant Identification System, where users can upload a leaf image and obtain a prediction result. The interface is designed to be simple and interactive, featuring an image upload option, a display area for the selected leaf, and a section showing the predicted plant name, confidence level and its benefits also. The interface also includes an AI chatbot that provides additional information such as the medicinal benefits and uses of the identified plant. The layout emphasizes ease of use, enabling even non-technical users to effectively interact with the system and receive clear identification feedback.

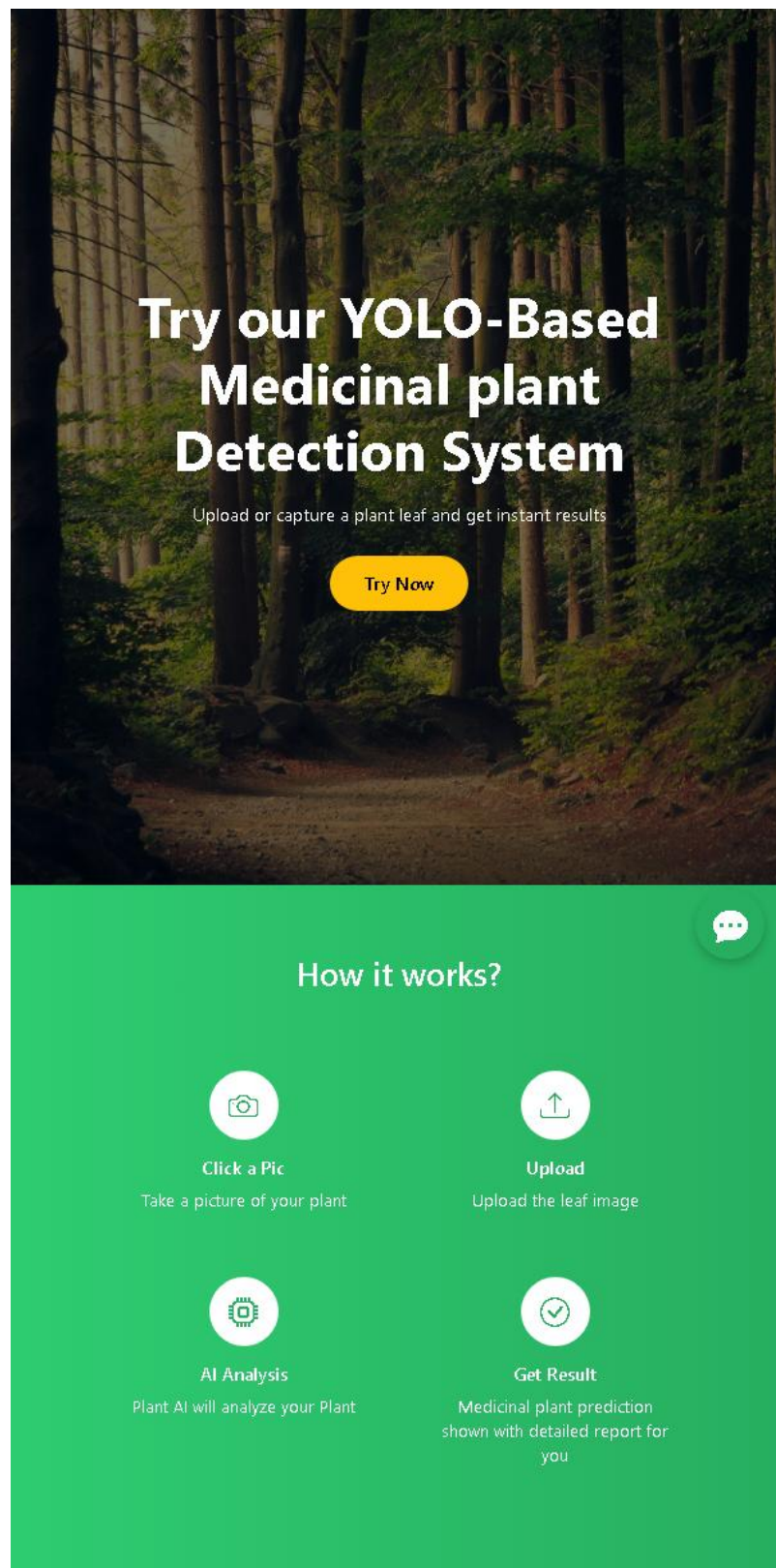


Figure 4.7: Interface for Uploading Leaf Image

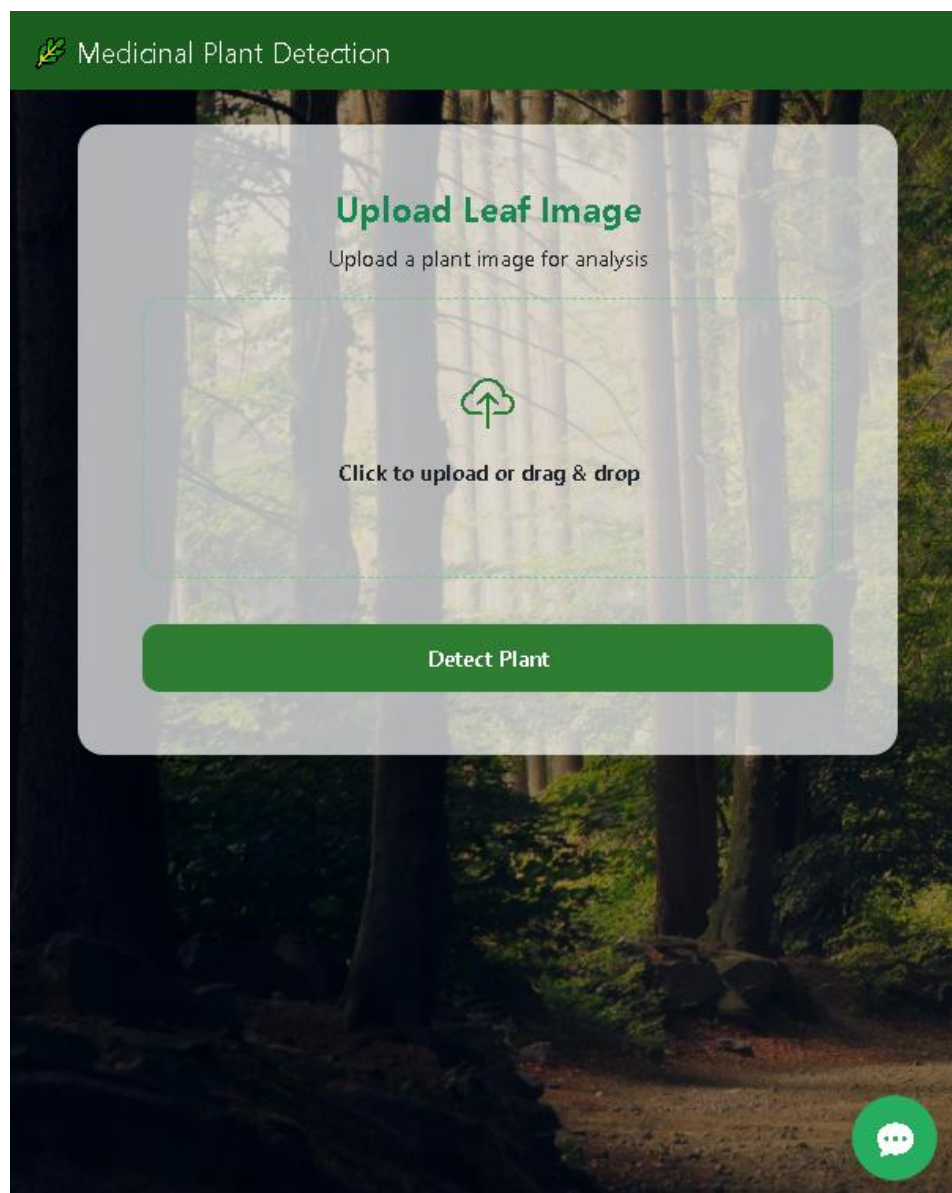


Figure 4.8: Interface to Upload a Plant image for analysis

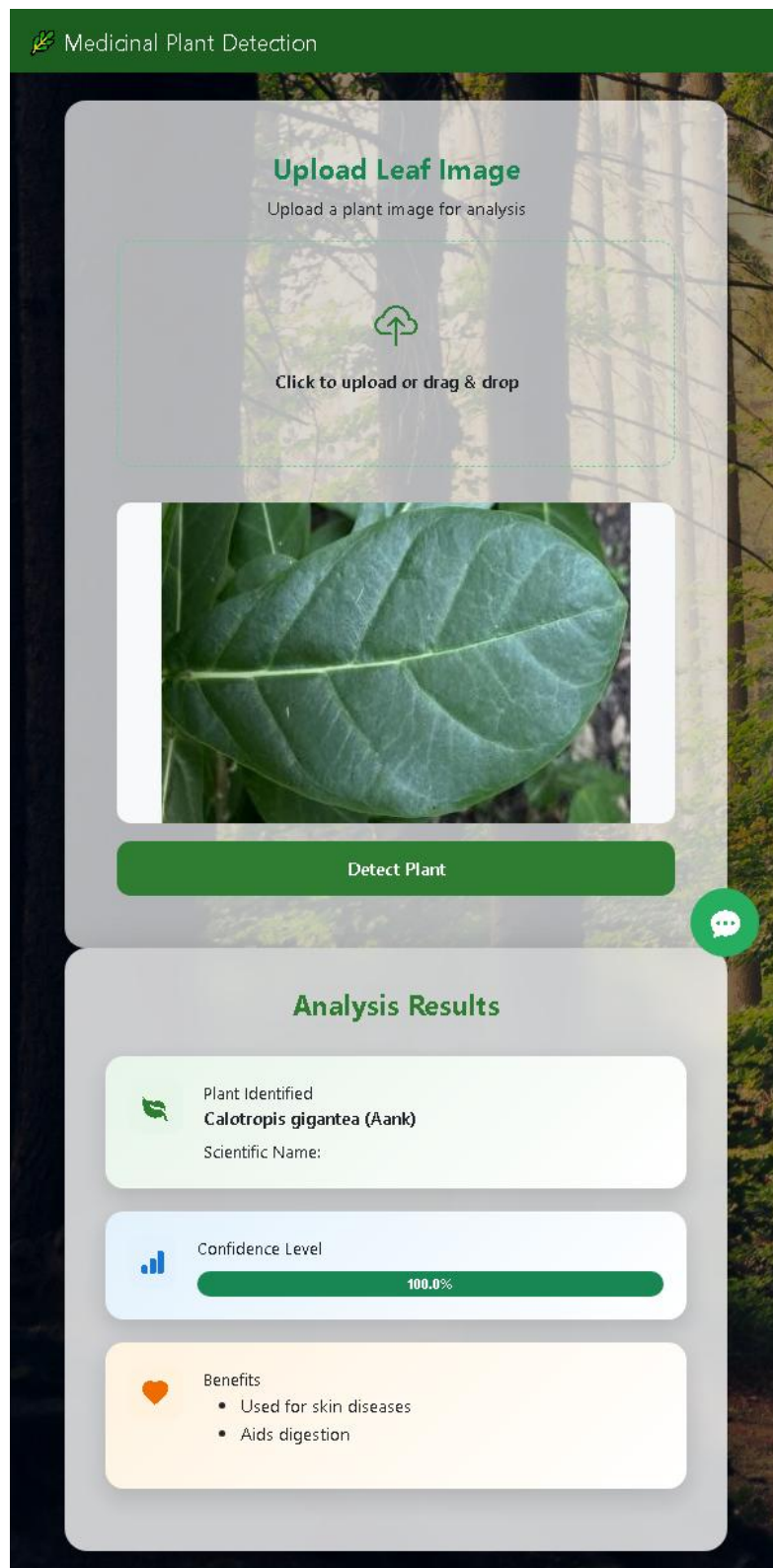


Figure 4.9: Interface for Displaying Prediction Result

CHAPTER 5: CONCLUSION & FUTURE WORK

5.1 Conclusion

Although the proposed Medicinal Plant Detection System has achieved promising performance using a YOLO-based deep learning model, several improvements and extensions remain for future development. The current system focuses mainly on identifying medicinal plants from leaf images; however, real-world plant identification can be more complex due to variations in lighting, background noise, leaf damage, and similarity between species. Future work will aim to enhance the dataset, improve the model's generalization, and expand the system's usability for broader audiences. Additionally, extending the system beyond leaf-based identification and incorporating recognition of other plant parts such as roots can improve accuracy and real-world reliability. These future enhancements will make the system more robust, scalable, and suitable for practical applications such as mobile-based plant identification and conservation support.

5.2 Future Work

1. **Increase Plant Species Coverage:** Expand the dataset by adding more medicinal plant species to improve real-world usability.
2. **Root-Based Plant Identification:** Extend the model to identify plants not only from leaves but also from roots and other visible plant parts.
3. **Improve Dataset Quality:** Collect more high-resolution images under different lighting, angles, and backgrounds to reduce bias and improve training.
4. **Mobile Application Deployment:** Develop a lightweight mobile app version of the system for real-time field usage.
5. **Real-Time Detection Support:** Integrate live camera-based detection so users can identify plants instantly without uploading images.
6. **Advanced Model Optimization:** Apply pruning, quantization, or smaller YOLO variants to improve speed and reduce model size for low-end devices.

7. **Better Handling of Similar Species:** Improve the model's ability to differentiate visually similar plants by adding more samples and using fine-grained training strategies.

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APPENDIX

The code and resources for the project are available at the following GitHub repository:

GitHub Repository link:

<https://github.com/apil-dotcom/Medicinal-plant-detection-system->

Collected Dataset link:

<https://www.kaggle.com/datasets/mirlab/medleaves-medicinal-plant-leaves-dataset>